LESSON PLAN

AIRCRAFT HUSBANDRY, CORROSION PREVENTION AND SURFACE PROTECTION (25)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introductions to various standards BS, AS, DTD, MIL, AN, NAS, close tolerance, unified etc.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introductions to various standards BS, AS, DTD, MIL, AN, NAS, close tolerance, unified etc.
5. Specific Objectives	25.1.1.1 Trainee will be able to describe the use of various standards, identification of bolts, nuts, safety methods and maintenance of cables.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of standard prefixes on sample fastener tags; Instructor supervision on correct identification of BS vs AN markings.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of using standardized fasteners in aircraft construction?
(b) Why is it critical to distinguish between British Standard (BS) and American National Standard (AN) fasteners?
(c) Can you list three common standard organizations used in aviation maintenance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and scope of BS standards • Overview of AS and DTD specifications • Identification of MIL and AN prefixes • Understanding NAS and unified thread standards • Significance of close tolerance markings	L & Ds	PPT, IB	15 min
(b)	• Comparison of thread series and pitch • Material designations across different standards • Safety implications of standard mismatches • Role of close tolerance in high-stress areas • Documentation requirements for standard usage	L & Ds	PPT, PDF	20 min
(c)	• Summary of key standard differences • Common pitfalls in standard identification • Best practices for maintenance documentation • Transition from legacy to modern standards • Importance of continuous standard updates	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between an AN fastener and a NAS fastener in terms of application.
(b) Why must a technician verify the standard type before installing a bolt in a critical structural area?
(c) How does the 'close tolerance' designation affect the selection of replacement fasteners?

12. Home Assignment and Follow Up	Prepare a brief note comparing the identification markings of BS and AN fasteners to be used in the next lesson on bolt identification.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Identification of AN-type, NAS etc.; grip length, torques, a/c bolt identification, screws, taper pins.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min